

SET

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The representation of an equilibrium system

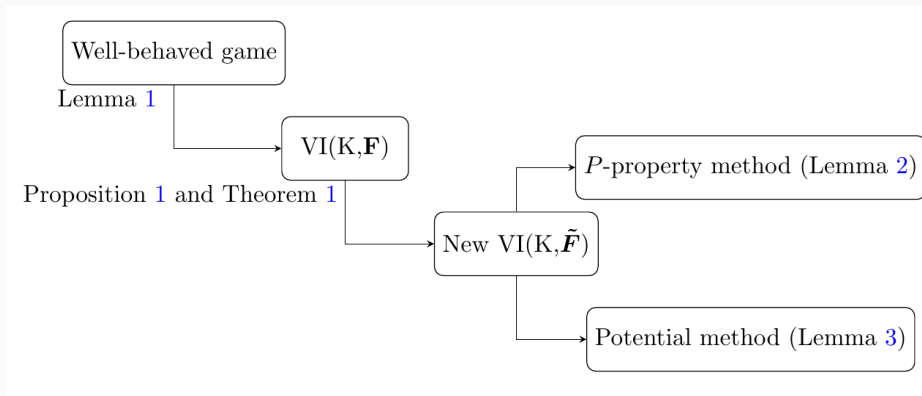
Many economic models reduce to nonlinear first-order conditions.

The difficulty

The original problem may obscure monotonicity, a potential function, or a useful uniqueness condition.

- Which transformations preserve the **entire equilibrium set**?
- Which representation makes the relevant structure easiest to verify?
- What can we learn about nonlinear network amplification?

The paper's map



An equilibrium is a best-response fixed point. The figure follows its VI representation through a sign-equivalent transformation to two methods for studying uniqueness.

Example: a two-player game

Two players choose $x_i \geq 0$, so $K = \mathbb{R}_+^2$.

Payoffs

$$\begin{aligned}u_1(x) &= x_1 \left(1 + \frac{1}{2}x_2 \right) - \frac{1}{2}x_1^2, \\u_2(x) &= 4 \left[x_2 \left(1 + \frac{1}{2}x_1 \right) - \frac{1}{2}x_2^2 \right].\end{aligned}$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = 1 + \frac{1}{2}x_2, \quad B_2(x_1) = 1 + \frac{1}{2}x_1.$$

The factor 4 changes player 2's payoff units, but leaves the best response unchanged.

Example: the VI and its transformation

For a candidate $x^* \in K$, use the negative marginal payoffs:

$$F_i = -\partial_i u_i, \quad F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ 4(x_2 - 1 - \frac{1}{2}x_1) \end{pmatrix}.$$

$$x^* \in \text{NE} \iff \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

SET: divide the second component by 4

$$\tilde{F}(x) = \text{diag}(1, \frac{1}{4})F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ x_2 - 1 - \frac{1}{2}x_1 \end{pmatrix}.$$

Positive scaling preserves every component's sign on K . Since K is rectangular,

$$\text{Sol}(K, F) = \text{Sol}(K, \tilde{F}).$$

Example: the two uniqueness routes

The transformed system has a symmetric derivative:

$$H = D\tilde{F} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}, \quad \lambda(H) \in \{\frac{1}{2}, \frac{3}{2}\}.$$

P-property method

The principal minors of H are
 $1, 1, \frac{3}{4} > 0$.

The P -matrix condition gives at most one VI solution.

Potential method

$$\begin{aligned} \Phi(x) &= \frac{1}{2}(x_1^2 + x_2^2) \\ &\quad - \frac{1}{2}x_1x_2 - x_1 - x_2. \end{aligned}$$

$\nabla\Phi = \tilde{F}$ and $\nabla^2\Phi = H \succ 0$. The VI minimizes the strictly convex Φ over K .

$$\tilde{F}(x^*) = 0 \quad \implies \quad \boxed{x_1^* = x_2^* = 2}.$$

This feasible zero proves existence. The equilibrium is **unique**.

Applications of general interest

1. **Market systems:** production networks, nonlinear peers, and oligopoly pricing.
2. **Nonlinear network complements:** when equilibrium exists, who acts most, and who is the key player?
3. **Network public goods:** how convex costs change participation, uniqueness, and effort rankings.

Only one methodological fact is needed before the examples.

The one methodological fact we need

Sign-equivalent transformation

If every component of F and $\tilde{F}$ has the same sign at every feasible profile, then on a product of individual intervals the two variational inequalities have **exactly the same solution set**.

Player's choice	Equilibrium condition	What matters
Lower bound	$F_i \geq 0$	Sign
Interior	$F_i = 0$	Zero
Upper bound	$F_i \leq 0$	Sign

The product-domain condition matters when a choice is at a boundary.

What a better representation can reveal

P -property route

A transformed derivative matrix can establish **at most one** equilibrium. **Used for:** production prices, peer effects, oligopoly pricing, CES complements.

Potential route

A symmetric transformed system can become an optimization problem; curvature can give uniqueness. **Used for:** network games, public goods, multiplex interactions.

Existence needs its own argument in each application.

CES production networks

- $a_{ij} \geq 0$ measures input dependence, with $\sum_j a_{ij} = 1 - \alpha_i$.
- Labor share $\alpha_i > 0$, productivity $z_i > 0$, wage $w > 0$.
- Input substitution elasticity $\sigma_i \geq 1$.

Equilibrium prices

For $0 < \alpha_i < 1$ and $\sigma_i > 1$, cost minimization gives

$$p_i z_i = C_i(p), \quad C_i(p) = w^{\alpha_i} \left(\frac{\sum_j a_{ij} p_j^{1-\sigma_i}}{1 - \alpha_i} \right)^{\frac{1-\alpha_i}{1-\sigma_i}}.$$

The Cobb–Douglas case $\sigma_i = 1$ follows by a limit.

Question: Does this nonlinear price system have exactly one solution?

Production networks: labor anchors the price system

A simple change of representation

From the nonlinear unit-cost equation $p_i z_i = C_i(p)$, take logarithms of both positive sides and set $x_i = \log(p_i/w)$.

$$D\tilde{F}(x) = I - W(x), \quad W(x) \geq 0, \quad \sum_j W_{ij}(x) = 1 - \alpha_i < 1.$$

- A positive labor share α_i limits the feedback through intermediate inputs.
- The log-price map is a contraction: the equilibrium price vector **exists and is unique**.
- Input linkages still transmit productivity shocks across industries.

Cobb–Douglas and the Leontief inverse

As $\sigma_i \rightarrow 1$, let $\varepsilon_i = \log z_i$ and $A = (a_{ij})$.

The nonlinear system becomes linear

$$x + \varepsilon - Ax = 0, \quad x^* = -(I - A)^{-1}\varepsilon.$$

$$L = (I - A)^{-1} = I + A + A^2 + \cdots, \quad \rho(A) \leq \max_i (1 - \alpha_i) < 1.$$

- $-\partial x_i / \partial \varepsilon_j = L_{ij}$ measures the price response to productivity in industry j .
- Powers of A collect direct and indirect input paths.
- The same resolvent $(I - \delta G)^{-1}$ appears in Bonacich centrality. Here the forcing term is $-\varepsilon$ and A contains input weights.

Existing application: nonlinear peer effects

A social norm need not be the average peer

$$\tilde{x}_{-i}(\beta) = \left(\sum_j \tilde{g}_{ij} x_j^\beta \right)^{1/\beta}, \quad x_i = (1 - \lambda_2) a_i + (\lambda_1 + \lambda_2) \tilde{x}_{-i}(\beta).$$

- $\beta = 1$ recovers linear-in-means; other values change whose behavior is most influential.
- Taking logs exposes a P -property: under the paper's parameter restrictions, **any solution is unique**.
- The result is conditional on existence; the transformation alone does not supply it.

Existing application: oligopoly with price bounds

Firm i chooses $p_i \in [a_i, b_i]$ and earns $(p_i - c_i)Q_i(p)$.

A transformed pricing condition

$$F_i = -\frac{\partial \pi_i}{\partial p_i} \sim_{\text{sign}} \tilde{F}_i = p_i - c_i + \frac{Q_i}{\partial Q_i / \partial p_i}.$$

The positive rescaling preserves the signs at both interior prices and binding price floors or caps.

- Under the paper's demand conditions E1–E3, the transformed system has the P -property.
- Therefore the Bertrand game has **at most one** pure-strategy equilibrium.

Existing applications: network games and multiplexity

Network public goods with income

An inverse Engel curve turns heterogeneous marginal conditions into a symmetric system on an undirected graph. A potential and a spectral condition can then give uniqueness.

Several network layers

An agent chooses effort across multiple kinds of ties. A transformed system can have a potential even with cross-layer costs. Extra curvature conditions are needed for uniqueness.

The same method can reveal different mathematical structures across models.

CES peer effects on an unnormalized network

$$u_i(x) = a_i x_i - \frac{1}{2} x_i^2 + \delta x_i \left(\sum_j g_{ij} x_j^\beta \right)^{1/\beta}.$$

Best response

$$T_i(x) = a_i + \delta S_i(x), \quad S_i(x) = \left(\sum_j g_{ij} x_j^\beta \right)^{1/\beta}.$$

- $a_i > 0$, $\delta > 0$, $\beta > 0$, and $x_i \geq 0$.
- G is an undirected adjacency matrix, without self-loops.
- We use a **connected, nonempty** network for the Perron-limit results.

What the CES parameter changes

For positive neighbor actions,

$$\frac{\partial S_i}{\partial x_j} = g_{ij} x_j^{\beta-1} S_i(x)^{1-\beta}.$$

For two linked neighbors j and k ,

$$\frac{\partial S_i / \partial x_j}{\partial S_i / \partial x_k} = \left(\frac{x_j}{x_k} \right)^{\beta-1}.$$

Relative peer influence

$\beta > 1$ favors high-action peers at the margin. $0 < \beta < 1$ places relatively more marginal weight on low-action peers.

Changing β also changes the scale of this unnormalized aggregate.

CES complements: uniqueness without a closed form

Every equilibrium has $x_i > 0$. Write $z_i = \log x_i$ and compare the positive quantities in the best-response equation:

$$H_i(z) = z_i - \log(a_i + \delta S_i(e^z)).$$

What the transformation reveals

$$DH(z) = I - W(z), \quad W(z) \geq 0, \quad \sum_j W_{ij}(z) = \frac{\delta S_i}{a_i + \delta S_i} < 1.$$

The positive baseline a_i leaves some own-action weight in every row. Hence there is **at most one** finite positive equilibrium.

Whether a finite equilibrium exists depends on the network threshold on the next slide.

The spectral existence threshold

Theorem 4

For $\bar{\delta} = \lambda_{\max}(G)^{-1/\beta}$:

- $0 < \delta < \bar{\delta}$: a unique positive equilibrium exists.
- $\delta \uparrow \bar{\delta}$: all actions diverge, with convergent ratios.
- $\delta \geq \bar{\delta}$: no finite positive equilibrium exists.

On a d -regular network with $a_i = a$,

$$x_i^* = \frac{a}{1 - \delta d^{1/\beta}}, \quad \bar{\delta} = d^{-1/\beta}.$$

Comparative statics below the threshold

How shocks change behavior

Within the region where a finite equilibrium exists,

$$a' \geq a \implies x^*(a', \delta, \beta) \geq x^*(a, \delta, \beta),$$

$$\delta' \geq \delta \implies x^*(a, \delta', \beta) \geq x^*(a, \delta, \beta).$$

- A higher individual baseline or stronger peer interaction raises equilibrium effort across the network.
- Equilibrium payoff is $u_i(x^*) = \frac{1}{2}(x_i^*)^2$: changes in effort also change welfare.

Bonacich centrality as the linear special case

When $\beta = 1$,

$$x = a + \delta Gx.$$

The network multiplier

$$(I - \delta G)x = a, \quad x^* = (I - \delta G)^{-1}a.$$

- $a = \mathbf{1}$: a standard Katz–Bonacich centrality vector.
- Heterogeneous a : baseline-weighted centrality.
- In the notation $(I - \delta G)x = b$, the vector b is this same baseline.

Eigenvector centrality at the critical limit

For the positive unit Perron vector e ,

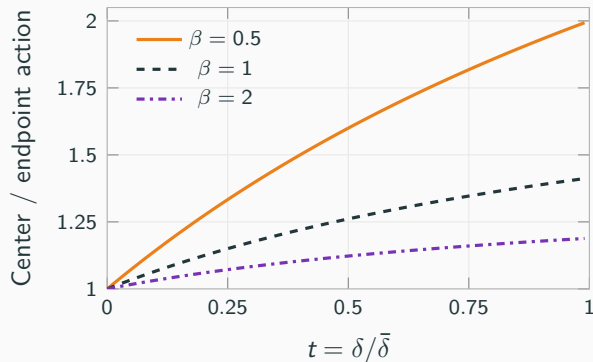
$$Ge = \lambda_{\max}(G)e, \quad e \gg 0.$$

Theorem 4(ii)

$$\lim_{\delta \uparrow \bar{\delta}} \frac{x_i^*}{x_j^*} = \left(\frac{e_i}{e_j} \right)^{1/\beta}.$$

- $\beta = 1$: $x^* / \|x^*\|_2 \rightarrow e$.
- General $\beta > 0$: the vector $(x^*)^\beta$ approaches the Perron direction.
- Baseline heterogeneity affects levels, but disappears from these limiting ratios.

Relative actions along the path



$$\frac{C}{L} = \frac{1 + tq}{1 + t/q}, \quad q = 2^{1/(2\beta)}.$$

As $t \uparrow 1$

The ratio converges to q .

The action levels diverge.

At the same fraction of the threshold, smaller β produces a larger center advantage here.

The key-player intervention

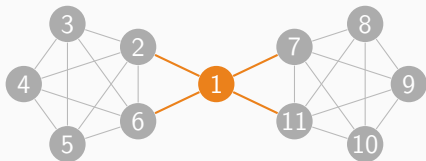
Let G^{-i} denote the network after removing player i .

Total equilibrium activity lost

$$\Delta_i = \sum_j x_j^*(G) - \sum_{j \neq i} x_j^*(G^{-i}), \quad i^* \in \arg \max_i \Delta_i.$$

- Removing a player eliminates their action and changes everyone else's incentives.
- Highest equilibrium action need not identify the largest total effect.
- For general β , recompute equilibrium on each remaining network.

A bridge can matter more than the most active player



Linear benchmark, $\beta = 1$

δ	Key-player type
0.1	Node 2
0.2	Node 1

Nodes 2, 6, 7, and 11 have the highest degree and the highest action.

Nonlinear peers change the intervention ranking

At $\delta = 0.1$, the paper finds a switch from node 1 to the node-2 type near $\beta = 0.722$ within its reported parameter range.

Network public goods with nonlinear costs

On an undirected network, player i chooses $x_i \geq 0$ and receives

$$u_i(x) = b\left(x_i + \delta \sum_j g_{ij}x_j\right) - c(x_i), \quad 0 < \delta \leq 1.$$

Assumption 2: benefits, costs, and a marginal crossing

$$\begin{aligned} b' > 0, \quad b'' < 0, \quad c'(x) > 0 \ (x > 0), \quad c'' \geq 0, \\ b'(0) > c'(0), \quad \lim_{x \rightarrow \infty} b'(x) < \lim_{x \rightarrow \infty} c'(x). \end{aligned}$$

The positive autarky effort k^* uniquely solves $b'(k^*) = c'(k^*)$.

The benchmark of Bramoullé and Kranton (2007) has $c(x) = x$ and $\delta = 1$.

Best responses, corners, and existence

Let $t_i = \delta \sum_j g_{ij} x_j$ be the public good supplied by neighbors.

If $x_i > 0$

The agent balances marginal benefit and cost:

$$b'(x_i + t_i) = c'(x_i).$$

If $x_i = 0$

Neighbors may already provide enough that contributing the first unit is not worthwhile.

More provision by neighbors weakly reduces one's best response. Every best response is at most the autarky effort k^* , so a finite equilibrium exists.

Substitutes: curvature can restore uniqueness

The original marginal conditions are asymmetric because each agent receives different total provision. A sign-equivalent inverse transformation gives

$$\tilde{F}_i(x) = x_i + \delta \sum_j g_{ij} x_j - h(x_i), \quad h = (b')^{-1} \circ c'.$$

Economic result

On an undirected graph this system has a potential. The sufficient condition

$$1 + \delta \lambda_{\min}(G) > 0$$

ensures a unique equilibrium. Convex private costs add curvature and can support uniqueness beyond this simple bound.

The sign equivalence also retains equilibria with zero effort.

Zero marginal cost at zero implies full participation

Now suppose c is strictly convex, $c'(0) = 0$, and $\delta = 1$, as in Proposition 7.

Every equilibrium is interior

At any finite profile with $x_i = 0$,

$$\left. \frac{\partial u_i}{\partial x_i} \right|_{x_i=0} = b' \left(\sum_j g_{ij} x_j \right) - c'(0) > 0.$$

Hence $x_i^* > 0$ for every player at every Nash equilibrium.

All equilibrium conditions therefore hold with equality:

$$b' \left(x_i^* + \sum_j g_{ij} x_j^* \right) = c'(x_i^*).$$

Full participation does not require equilibrium uniqueness.

Complete networks: convex costs remove the continuum

Take the complete network and $\delta = 1$. Every player receives total provision $S = \sum_j x_j$.

Strictly convex costs, $c'(0) = 0$

$$c'(x_i) = b'(S) \quad \text{for all } i.$$

Strictly increasing c' forces equal effort.

$$x_i = \bar{x}, \quad b'(n\bar{x}) = c'(\bar{x}).$$

The equilibrium is unique.

Linear costs, $c(x) = x$

With $b'(k^*) = 1$, every profile satisfying

$$x_i \geq 0, \quad \sum_i x_i = k^*$$

is an equilibrium.

The continuum includes both equal sharing and a single specialist.

Here $\lambda_{\min}(G) = -1$. Uniqueness with strictly convex costs holds even though $1 + \delta\lambda_{\min}(G) = 0$.

Network structure determines effort rankings

Maintain homogeneous b, c , strictly convex costs, $c'(0) = 0$, and $\delta = 1$. Let N_i and $\bar{N}_i = N_i \cup \{i\}$ denote open and closed neighborhoods.

Neighborhood inclusion

At every equilibrium,

$$N_i \subsetneq N_j \text{ or } \bar{N}_i \subsetneq \bar{N}_j \implies x_i^* > x_j^*.$$

- **Star**, $n \geq 3$. Leaves choose equal effort. The hub contributes less.
- **Nested split graph**. If $d_i < d_j$, then $x_i^* > x_j^*$.

Degree alone does not imply this ranking on arbitrary networks.

What SET establishes in these applications

Question	Required argument
Equilibrium equivalence	Valid VI representation, full-domain sign equivalence, and appropriate domain geometry.
Existence	A compactness, coercivity, strong-P, or bounded-iteration argument.
Uniqueness	A suitable P property, strict monotonicity, or strict convexity.

Main conclusions

A general method

On rectangular domains, the signs of the equilibrium operator determine the VI solution set. Suitable transformations can reveal structure absent from the original Jacobian.

- CES complements combine a SET uniqueness argument with a Perron existence threshold.
- Bonacich centrality is a resolvent response. Eigenvector centrality describes its critical direction.
- Under strategic substitutes, cost curvature and the smallest network eigenvalue govern uniqueness.

Which transformation exposes the useful structure in a given model?

Backup: a formal coordinate proof of Theorem 1

Let $K = \prod_i K_i$ and suppose $x^* \in \text{Sol}(K, F)$. For every i and $t \in K_i$,

$$(t - x_i^*)F_i(x^*) \geq 0.$$

Sign equivalence implies the same inequality for $\tilde{F}_i(x^*)$:

$$\begin{cases} F_i(x^*) = 0 & \implies \tilde{F}_i(x^*) = 0, \\ F_i(x^*) > 0 & \implies t - x_i^* \geq 0 \text{ and } \tilde{F}_i(x^*) > 0, \\ F_i(x^*) < 0 & \implies t - x_i^* \leq 0 \text{ and } \tilde{F}_i(x^*) < 0. \end{cases}$$

Consequently, for every $x \in K$,

$$\langle x - x^*, \tilde{F}(x^*) \rangle = \sum_i (x_i - x_i^*) \tilde{F}_i(x^*) \geq 0.$$

Thus $\text{Sol}(K, F) \subseteq \text{Sol}(K, \tilde{F})$. Interchanging the two operators gives the reverse inclusion.

Backup: matrix tests in the payoff-scaling example

For $J = \begin{pmatrix} 1 & -\delta \\ -\kappa\delta & \kappa \end{pmatrix}$ with $\kappa > 0$:

P-matrix test

The principal minors are

$$1, \quad \kappa, \quad \kappa(1 - \delta^2).$$

All are positive if $0 < \delta < 1$.

Strict monotonicity test

For $S = (J + J^\top)/2$,

$$\det S = \kappa - \frac{(1 + \kappa)^2 \delta^2}{4}.$$

$S \succ 0$ iff $\delta < 2\sqrt{\kappa}/(1 + \kappa)$.

Let $\Delta = (1 - \kappa)^2 + (1 + \kappa)^2 \delta^2$. The smallest eigenvalue is

$$\lambda_{\min}(S) = \frac{1 + \kappa - \sqrt{\Delta}}{2}.$$

Rescaling by $\text{diag}(1, 1/\kappa)$ restores the symmetric derivative with eigenvalues $1 - \delta$ and $1 + \delta$.

Backup: the nonlinear Perron limit

Let $y(\delta) = x(\delta)^\beta$ and normalize $q(\delta) = y(\delta)/\|y(\delta)\|_1$.

- If actions remained bounded as $\delta \uparrow \bar{\delta}$, continuity would give an equilibrium at $\bar{\delta}$.
- Connectedness propagates divergence: $x_i \geq \delta x_j$ whenever $g_{ij} = 1$.

Therefore $a_i/x_i \rightarrow 0$, and along any convergent subsequence,

$$\frac{(x - a)^\beta}{\|x^\beta\|_1} = \delta^\beta Gq(\delta) \implies q = \bar{\delta}^\beta Gq.$$

Uniqueness of the normalized limit

The only nonnegative unit- ℓ_1 Perron vector is $e/\|e\|_1$. Every subsequential limit is the same, so the full sequence converges.

Backup: algebraic solutions and positive equilibria

For two linked agents with baseline $a = \mathbf{1}$,

$$G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda_1 = 1, \quad \lambda_2 = -1.$$

Whenever $\delta \neq \pm 1$, the linear system has the solution

$$x = (I - \delta G)^{-1} \mathbf{1} = \frac{1}{1 - \delta} \mathbf{1}.$$

Parameter	Algebraic system	Economic equilibrium
$0 < \delta < 1$	Unique solution	Positive and finite
$\delta = 1$	Singular, inconsistent	None
$\delta > 1$	Unique negative solution	None on $\mathbb{R}_+^2$

Backup: complete-network uniqueness without the strict spectral bound

Assume homogeneous b, c , strictly convex c , $c'(0) = 0$, and $\delta = 1$. On the complete network, write $S = \sum_j x_j$.

1. Full participation makes every first-order condition an equality:

$$c'(x_i) = b'(S) = c'(x_j) \quad \text{for every } i, j.$$

2. Strictly increasing c' implies $x_i = x_j = \bar{x}$.
3. The function $f(z) = b'(nz) - c'(z)$ is strictly decreasing, with $f(0) > 0$ and an eventually negative value. Thus it has exactly one positive zero.

Why the spectral test is only sufficient

For $n \geq 2$, the complete graph has $\lambda_{\min}(G) = -1$. At $\delta = 1$ the strict spectral inequality fails, but the argument above proves uniqueness.

Backup: why the hub contributes less in a star

Let $n \geq 3$ and $\delta = 1$, with homogeneous functions, strictly convex costs, and $c'(0) = 0$. Write x_H for the hub's effort.

1. Every leaf faces the same peer effort x_H and has a unique best response. Thus all leaves choose one common effort $x_L > 0$.
2. The hub and a leaf satisfy

$$c'(x_H) = b'(x_H + (n-1)x_L), \quad c'(x_L) = b'(x_H + x_L).$$

3. Since $(n-2)x_L > 0$ and b' is strictly decreasing,

$$c'(x_H) < c'(x_L).$$

Strictly increasing c' then gives $x_H < x_L$.

The proof applies to every equilibrium, without a uniqueness assumption.

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The seminar follows the supplied SET draft. The two-player walkthrough, path-network curves, and two-node calculation are seminar examples.